

Generation Interconnection Request System Impact Study Report

For Queue Project Q21-1810

Murdock 230 kV
July 20, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1810. The Interconnection Customer proposes a 244.1 MW solar photovoltaic facility interconnecting at the Murdock 230 kV substation in the Indiana area. The request was received on July 09, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 629 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$3,534,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
East Methuen - Henry County 500 kV	E Methuen - Lonetree 500 kV	100.0	2,789	33.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
East Methuen - Henry County 500 kV	Reconductor line	\$76,695,000	5% (\$3,534,000)
	Total allocated to project		\$3,534,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.